

MIAE 221 · Rapid Review Sheet · Part 4

Crystallographic Densities, XRD, Anisotropy & Bragg's Law

1. Linear & Planar Atomic Densities

- Linear Density (LD):** Number of atom diameters centered on a direction vector per unit length:

$$LD = \frac{n}{L}$$

- $FCC [100]$: $L = a = 2\sqrt{2}R$, $n = 2(\frac{1}{2}) = 1 \implies LD_{[100]} = \frac{1}{2\sqrt{2}R}$.
- $FCC [110]$ (Close-Packed): $L = 4R$, $n = 2 \implies LD_{[110]} = \frac{2}{4R} = \frac{1}{2R}$.
- Planar Density (PD):** Number of atoms centered on a crystallographic plane per unit area of that plane:

$$PD = \frac{n}{A}$$

- $FCC (110)$: Area $A = a \cdot a\sqrt{2} = 8\sqrt{2}R^2$; Atoms $n = 4(\frac{1}{4}) + 2(\frac{1}{2}) = 2 \implies PD_{(110)} = \frac{2}{8\sqrt{2}R^2} = \frac{1}{4\sqrt{2}R^2}$.
- $FCC (111)$ (Close-Packed): Area $A = \frac{\sqrt{3}}{2}a^2\sqrt{2} = 4\sqrt{3}R^2$; Atoms $n = 3(\frac{1}{6}) + 3(\frac{1}{2}) = 2 \implies PD_{(111)} = \frac{2}{4\sqrt{3}R^2} = \frac{1}{2\sqrt{3}R^2} \approx \frac{0.29}{R^2}$ (Maximum planar density in FCC).
- Slip System Law:** Plastic deformation occurs along directions of **highest LD** within planes of **highest PD** .

2. Single Crystal Anisotropy vs. Polycrystalline Isotropy

- Single Crystal Anisotropy:** Properties depend strongly on orientation:
 - BCC Iron: $E_{[111]} = 272.7$ GPa (close-packed, stiffest) vs. $E_{[100]} = 125.0$ GPa (most compliant) $\implies$ Ratio ≈ 2.18 .
 - Single-crystal turbine blades (CMSX-4) eliminate high-temperature grain-boundary creep along $[001]$.
- Polycrystalline Isotropy:** Millions of randomly oriented microscopic grains average out directional stiffness, yielding macroscopic isotropy.

3. Interplanar Spacing & Bragg's Law

- Cubic Interplanar Spacing (d_{hkl}):**

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

- Bragg's Law for Constructive Interference:**

$$n\lambda = 2d_{hkl} \sin \theta$$

- λ = X-ray wavelength (typically 0.15418 nm for $Cu-K_\alpha$ or 0.1790 nm for $Fe-K_\alpha$).
- d_{hkl} = Spacing between parallel (hkl) planes.
- θ = Bragg angle (half of the detector angle 2θ).
- n = Order of reflection (usually $n = 1$).

4. Diffractometry & Systematic Reflection Conditions

Crystal Structure	Diffraction Selection Rule	First 4 Diffraction Peaks
BCC (Body-Centered)	$h + k + l = \text{Even}$	(110), (200), (211), (220)
FCC (Face-Centered)	h, k, l unmixed (all even or all odd)	(111), (200), (220), (311)

- Exam Tip:* If the first peak has $h^2 + k^2 + l^2 = 2 \implies (110) \implies$ **BCC**. If $h^2 + k^2 + l^2 = 3 \implies (111) \implies$ **FCC**.

5. Polymorphism & Allotropy

- Polymorphism:** Existence of > 1 crystal structure depending on T and P . In elements: **Allotropy**.
- Carbon:** Graphite (sp^2 2D sheets, van der Waals, conductive lubricant) vs. Diamond (sp^3 tetrahedral network, hardest, insulator).
- Iron (Fe):** α -Fe (BCC, $< 912^\circ\text{C}$) $\rightarrow$ γ -Fe (FCC, $912 - 1394^\circ\text{C}$) $\rightarrow$ δ -Fe (BCC, $> 1394^\circ\text{C}$).
 - Volume Contraction:* Heating through 912°C causes iron to contract because FCC (APF = 0.74) is denser than BCC (APF = 0.68).